Development and Internal Validation of an Explainable Clinical Decision Support System for Predicting Micro-TESE Outcomes in Non-Obstructive Azoospermia: A Mul Agent AI-Assisted Approach

Authors: Hossein Jamalirad, Marjan Sabbaghian, Hassan Vakili Arki

Abstract

Background: Non-obstructive azoospermia (NOA) is a major cause of male infertility, and micro-TESE outcomes remain difficult to predict preoperatively. Methods: We analyzed 2,413 NOA patients from the Royan Institute Andrology Clinic (MUMS), using a 45-feature dataset (37 numeric + 8 categorical) that included 18 bilateral pathology features extracted from free-text reports. Sixteen machine-learning models were benchmarked, and five were retained in the final v2 pipeline. Results: CatBoost achieved the best performance with AUC 0.8306 (95% CI 0.823–0.845), accuracy 0.770, sensitivity 0.678, specificity 0.823, and F1 0.684. Conclusion: Pathology-integrated modeling improved predictive performance and supports explainable CDSS development for NOA micro-TESE counseling.

Key Results Summary

Authoritative v2 results: 2,413 patients, 45 features, and 18 pathology features. Sixteen models were tested and five finalized. CatBoost was best (AUC 0.8306; 95% CI 0.823–0.845).

Tables

1. Table 1. Patient Demographics and Clinical Characteristics (N=2,413)

Headers: Characteristic | Value
- Total patients | 2,413
- Successful sperm retrieval, n (%) | 886 (36.7%)
- Unsuccessful sperm retrieval, n (%) | 1,527 (63.3%)
- Prevalence of successful retrieval | 0.367
- Class imbalance ratio (failure:success) | 1.72:1
- Primary micro-TESE, n (%) | 2,203 (91.2%)
- Success rate | 32.5%
- Salvage micro-TESE, n (%) | 210 (8.8%)
- Success rate | 80.5%
- Ethnicity: Iranian, n (%) | 2,100 (87.0%)
- Ethnicity: Afghan, n (%) | 251 (10.4%)
- Ethnicity: Iraqi, n (%) | 48 (2.0%)
- FSH — missing | 22.2%
- LH — missing | 25.1%
- Testosterone — missing | 22.6%
- Prolactin — missing | 40.4%
- Estradiol — missing | 41.6%
- Karyotype — missing | 62.3%
- Y chromosome microdeletion — missing | 62.4%
- BMI — missing | 64.1%
- Total features | 45 (37 numeric + 8 categorical)
- Pathology features (bilateral RT/LT) | 18
- Models tested | 16
- Finalized models in v2 | 5

2. Table 2. Model Performance Comparison Using 5×5 Nested Cross-Validation

Headers: Model | AUC (95% CI) | Accuracy | Sensitivity | Specificity | F1 | PPV | NPV | MCC
- CatBoost* | 0.8306 (0.823–0.845) | 0.770 | 0.678 | 0.823 | 0.684 | — | — | —
- XGBoost | 0.8214 (0.814–0.835) | 0.758 | 0.669 | 0.809 | 0.669 | — | — | —
- LightGBM | 0.8213 (0.809–0.836) | 0.755 | 0.681 | 0.798 | 0.671 | — | — | —

- RandomForest | 0.8157 (0.809–0.829) | 0.747 | 0.694 | 0.778 | 0.669 | — | — | —
- Logistic Reg. | 0.7926 (0.777–0.804) | 0.735 | 0.659 | 0.779 | 0.646 | — | — | —

3. Table 3. Calibration Assessment: 10-Bin Quantile Calibration of CatBoost

Headers: Bin | Mean Predicted Probability | Observed Fraction Positive | |Deviation|
- 1 | 0.104 | 0.058 | 0.046
- 2 | 0.181 | 0.100 | 0.081
- 3 | 0.249 | 0.154 | 0.095
- 4 | 0.298 | 0.174 | 0.124
- 5 | 0.346 | 0.264 | 0.082
- 6 | 0.407 | 0.320 | 0.087
- 7 | 0.483 | 0.336 | 0.147
- 8 | 0.580 | 0.560 | 0.020
- 9 | 0.786 | 0.780 | 0.006
- 10 | 0.904 | 0.926 | 0.022
- | | Mean |deviation| | 0.071

4. Table 4. Top 15 Predictive Features Ranked by Mean Absolute SHAP Value

Headers: Rank | Feature | Mean |SHAP| | Category | Direction
- 1 | RT Severe Hypospermatogenesis % | 0.3958 | Pathology | Higher → ↑ retrieval
- 2 | RT Hypospermatogenesis % | 0.2007 | Pathology | Higher → ↑ retrieval
- 3 | LT Severe Hypospermatogenesis % | 0.1590 | Pathology | Higher → ↑ retrieval
- 4 | RT MA Spermatocytic % | 0.1473 | Pathology | Context-dependent
- 5 | LT Severity Score | 0.1308 | Pathology (ordinal) | Higher → ↑ retrieval
- 6 | LT SCO % | 0.1099 | Pathology | Higher → ↓ retrieval
- 7 | RT SCO % | 0.0898 | Pathology | Higher → ↓ retrieval
- 8 | RT Severity Score | 0.0855 | Pathology (ordinal) | Higher → ↑ retrieval
- 9 | RT MA Elongated % | 0.0672 | Pathology | Higher → ↑ retrieval
- 10 | Habits | 0.0663 | Lifestyle | Context-dependent
- 11 | Klinefelter Status | 0.0612 | Genetic | Present → ↓ retrieval
- 12 | LT MA Spermatocytic % | 0.0606 | Pathology | Context-dependent
- 13 | LT Hypospermatogenesis % | 0.0565 | Pathology | Higher → ↑ retrieval
- 14 | Seminal Plasma pH | 0.0550 | Semen | Context-dependent
- 15 | RT Testis Volume (Sono) | 0.0501 | Testicular | Higher → ↑ retrieval

5. Table 5. Multi-Agent Verification Audit: Critical Bugs Identified and Corrected

Headers: # | Bug Description (v1) | Correction Applied (v2) | Severity
- 1 | IQR clipper collapsed all 16 pathology features to constant 0 (IQR=0) | IQR clipper removed; raw pathology percentages + severity scores retained | Critical
- 2 | Nested CV = 2×2 with 2 iterations (4 fits/fold; underpowered) | 5×5 nested CV with 50 iterations (250 fits/fold) | Critical
- 3 | Representative params selected by argmax of outer-test AUC (data leakage) | Selected by median of inner-CV best score across folds | Critical
- 4 | Height & Weight included alongside BMI (collinearity) | Height & Weight dropped; BMI retained | High
- 5 | Hormone columns not coerced to numeric before pipeline | pd.to_numeric(errors="coerce") applied immediately after loading | High
- 6 | Preprocessing: KNNImputer → IQR clipper → StandardScaler | StandardScaler → KNNImputer (no IQR clipper) | High
- 7 | Hard-coded decision threshold of 0.5 | Youden's J on inner-CV out-of-fold predictions | High
- 8 | Fairness analysis on resubstitution predictions | Fairness computed on out-of-fold predictions only | High
- 9 | Race included as model feature | Race excluded; used only as protected attribute for fairness | High

Figures

Figure 1: CDSS system architecture showing the stateless web application with PHP 8.0 backend, CatBoost model inference, and SHAP-based explainability output.

Figure 2: Model performance comparison across 5 classifiers (CatBoost, XGBoost, LightGBM, Random Forest, Logistic Regression) showing AUC with 95% CI.

Figure 3: SHAP summary plot showing top predictive features with beeswarm visualization of feature contributions.

Figure 4: Calibration plot showing 10-bin quantile calibration of CatBoost with systematic SMOTE overestimation in low-mid probability bins.

Figure 5: Decision curve analysis showing CatBoost net benefit exceeding treat-all and treat-none strategies across threshold probabilities 0.05-0.60.

Figure 6: Multi-agent verification pipeline architecture and bug discovery timeline showing v1 to v2 improvement.